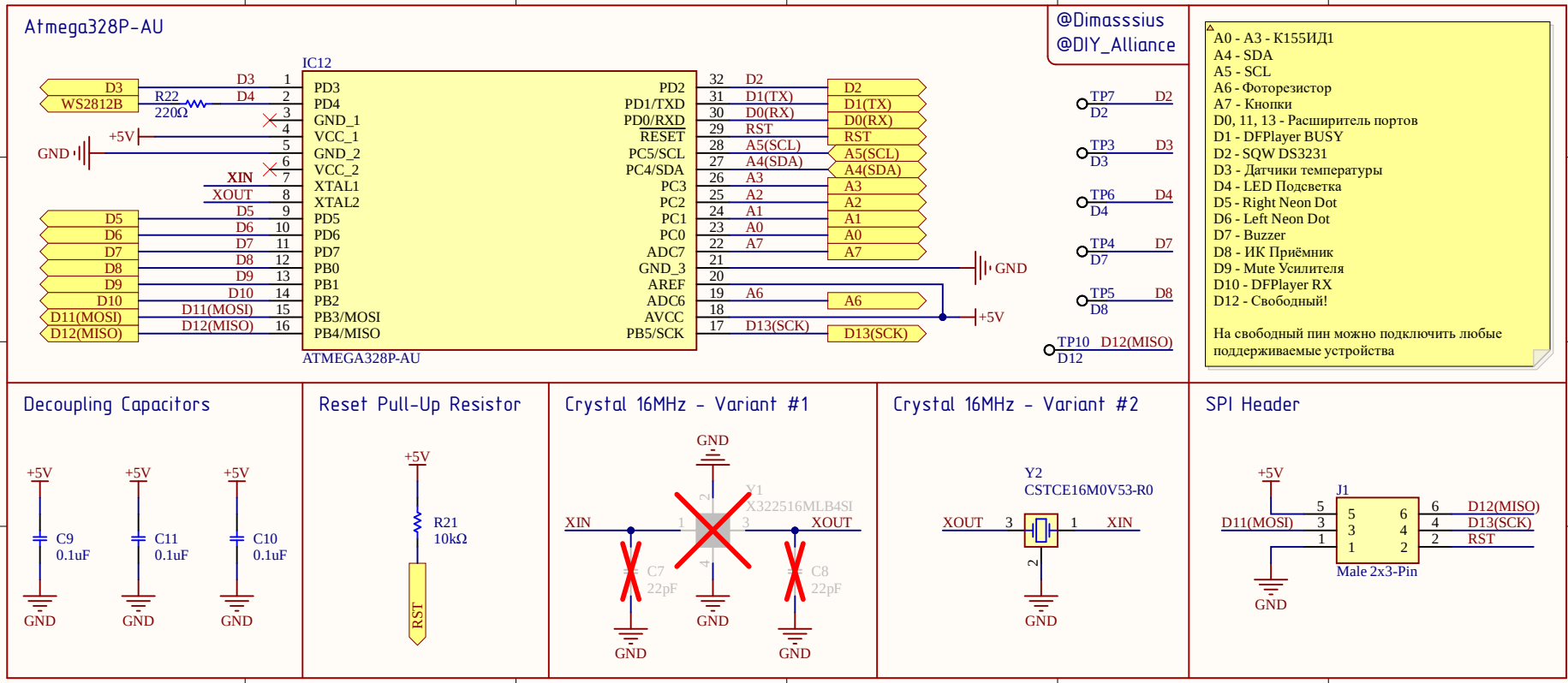
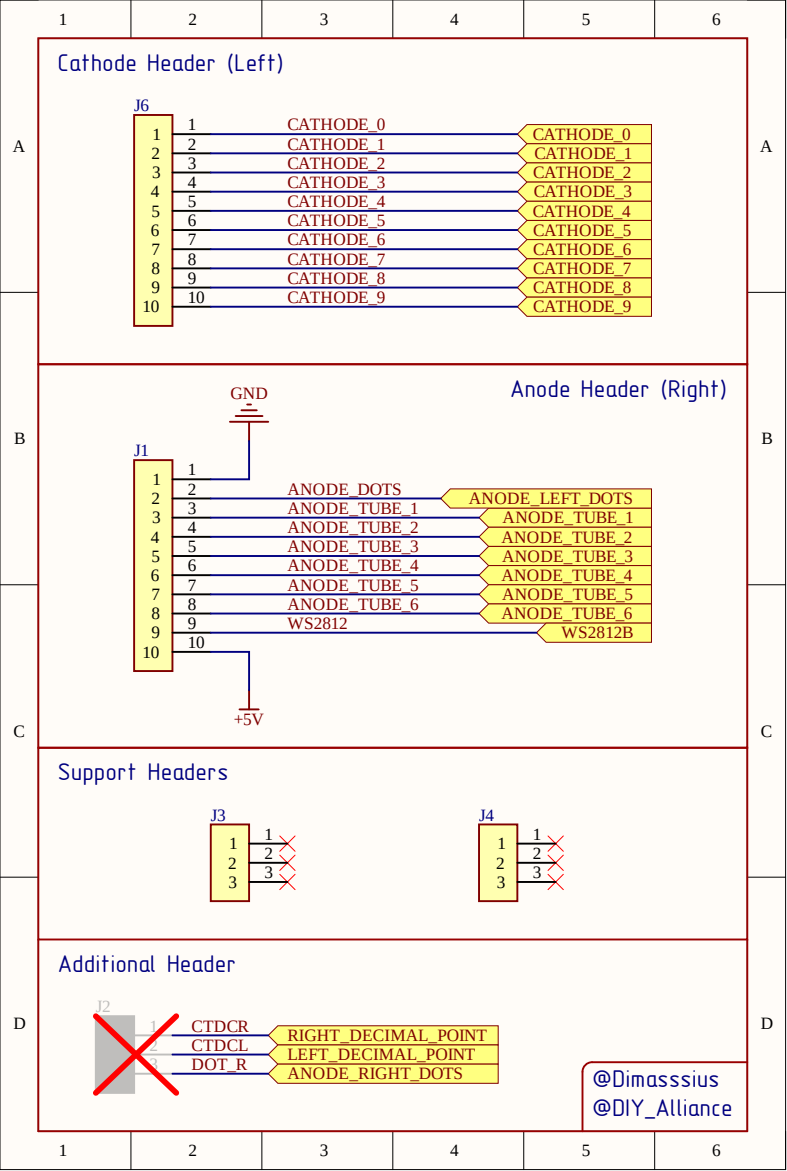
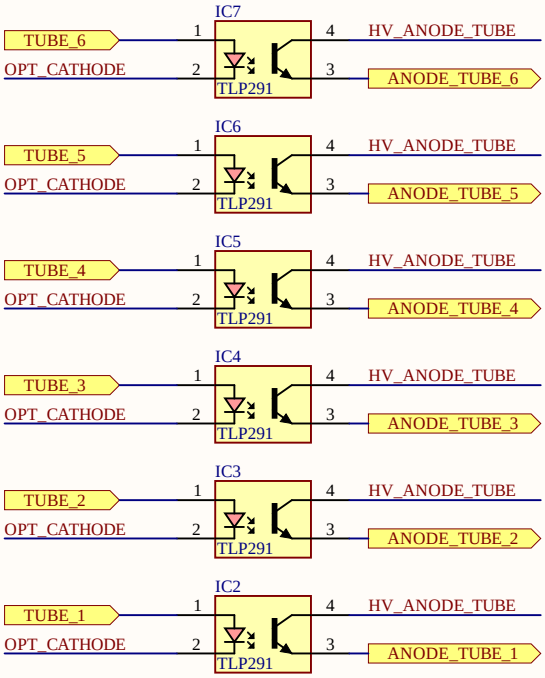
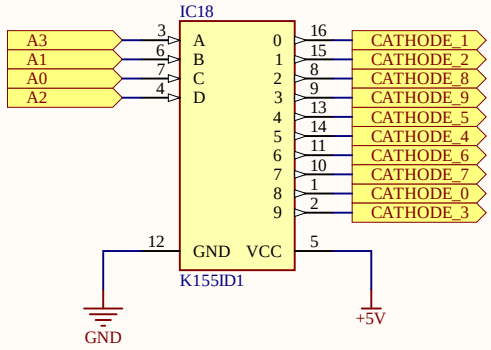

Atmega328P-AU
IC12
ATMEGA328P-AU
@Dimasssius
@DIY_Alliance
A0 - A3 - К155ИД1
A4 - SDA
A5 - SCL
A6 - Фоторезистор
A7 - Кнопки
D0, 11, 13 - Расширитель портов
D1 - DFPlayer BUSY
D2 - SQW DS3231
D3 - Датчики температуры
D4 - LED Подсветка
D5 - Right Neon Dot
D6 - Left Neon Dot
D7 - Buzzer
D8 - ИК Приёмник
D9 - Mute Усилителя
D10 - DFPlayer RX
D12 - Свободный!
На свободный пин можно подключить любые поддерживаемые устройства
Decoupling Capacitors
C9 0.1uF
C11 0.1uF
C10 0.1uF
Reset Pull-Up Resistor
R21 10kΩ
RST
Crystal 16MHz - Variant #1
Y1 X322516MLB4SI
C7 22pF
C8 22pF
Crystal 16MHz - Variant #2
Y2 CSTCE16M0V53-R0
SPI Header
J1
Male 2x3-Pin


Cathode Header (Left)
J6
1 CATHODE_0
2 CATHODE_1
3 CATHODE_2
4 CATHODE_3
5 CATHODE_4
6 CATHODE_5
7 CATHODE_6
8 CATHODE_7
9 CATHODE_8
10 CATHODE_9
CATHODE_0
CATHODE_1
CATHODE_2
CATHODE_3
CATHODE_4
CATHODE_5
CATHODE_6
CATHODE_7
CATHODE_8
CATHODE_9
Anode Header (Right)
GND
J1
1
2 ANODE_DOTS
3 ANODE_TUBE_1
4 ANODE_TUBE_2
5 ANODE_TUBE_3
6 ANODE_TUBE_4
7 ANODE_TUBE_5
8 ANODE_TUBE_6
9 WS2812
10
+5V
ANODE_LEFT_DOTS
ANODE_TUBE_1
ANODE_TUBE_2
ANODE_TUBE_3
ANODE_TUBE_4
ANODE_TUBE_5
ANODE_TUBE_6
WS2812B
Support Headers
J3
J4
Additional Header
J2
CTDCR
CTDCL
DOT_R
RIGHT_DECIMAL_POINT
LEFT_DECIMAL_POINT
ANODE_RIGHT_DOTS
@Dimasssius
@DIY_Alliance

Anode Tubes Drivers



Decoder - K155ID1



Anode Tubes Resistor

Данным резистором можно регулировать яркость свечения индикаторов и разделительных точек, чем меньше сопротивление, тем выше яркость (и выше ток, что влияет на срок жизни индикаторов)

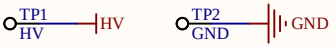
Варианты:
1. 33 кОм
2. 10 кОм (рекомендуемый)
3. 4.7 кОм
4. 3.3 кОм



Optoisolators Cathode Resistor

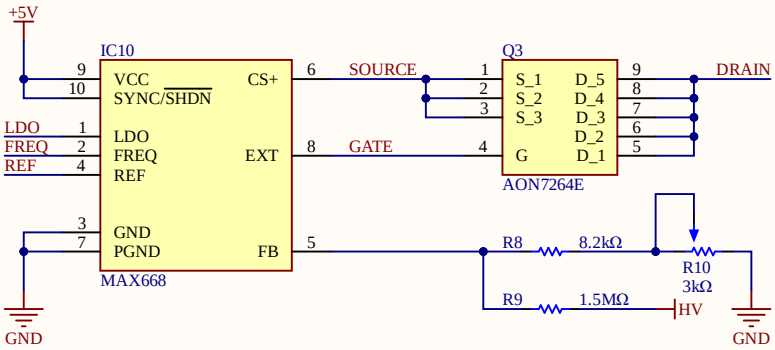


Test Point - High Voltage

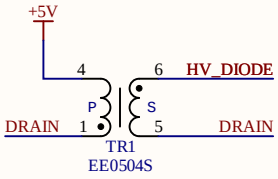


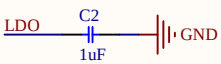
@Dimasssius
@DIY_Alliance

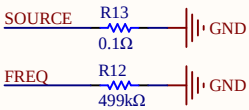
High Voltage Generator

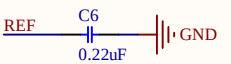


Transformer 1:10

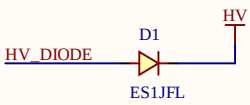


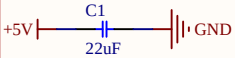


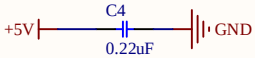


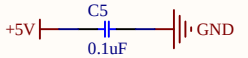


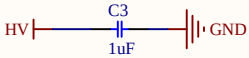
+180V



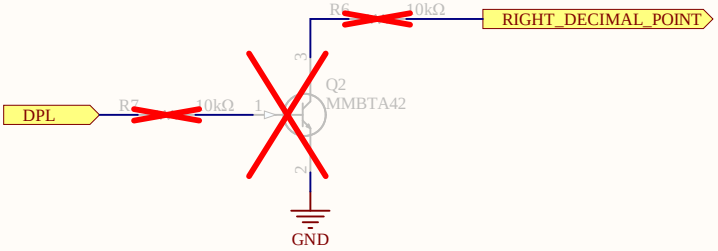


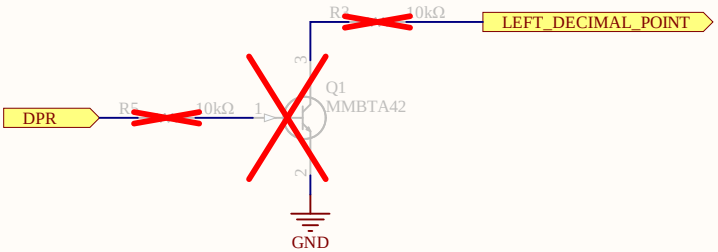




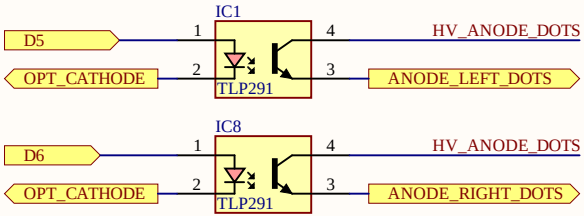


Nixie Tube Dots





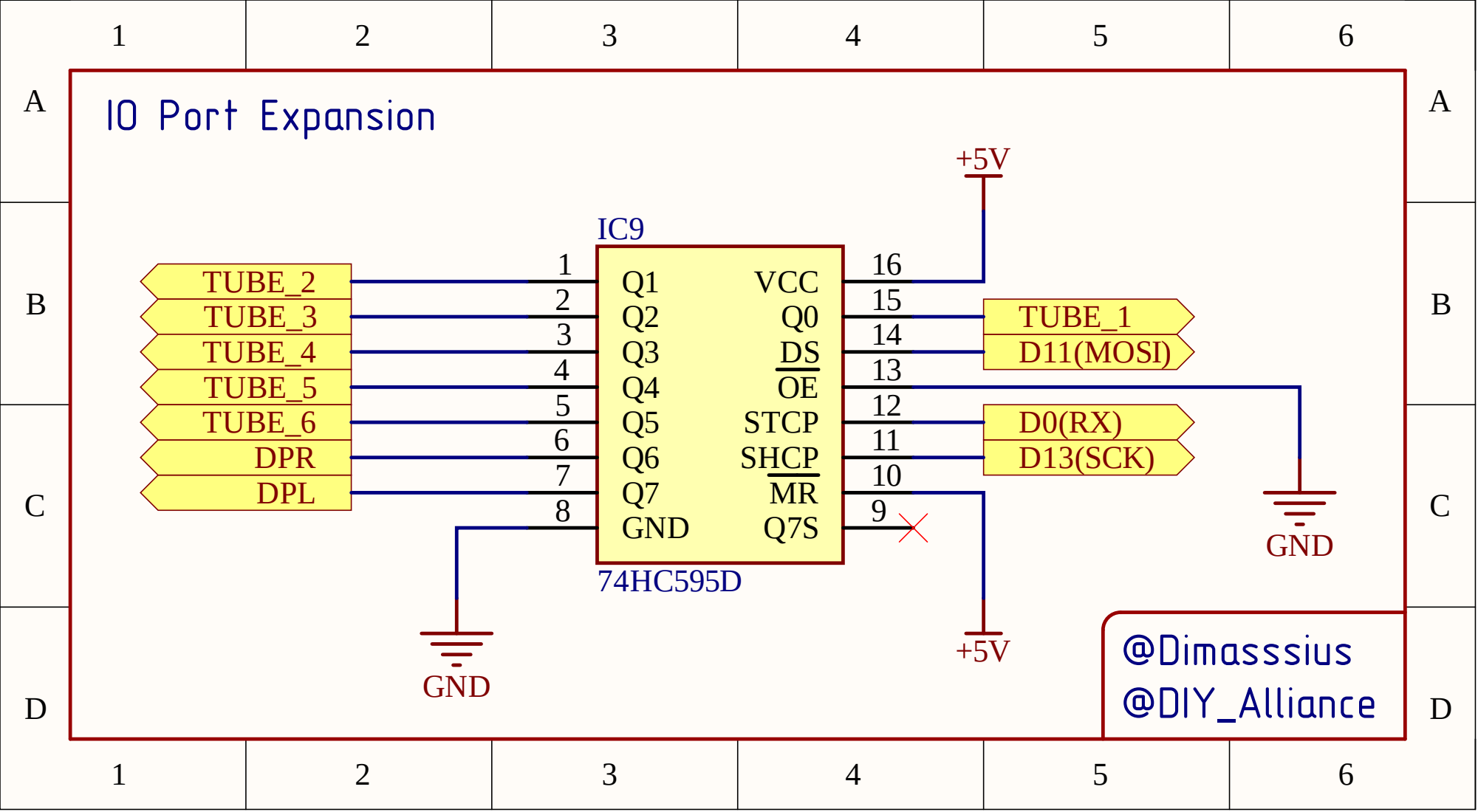
Neon Dots (NE-2, ИН-3, ИНС-1)

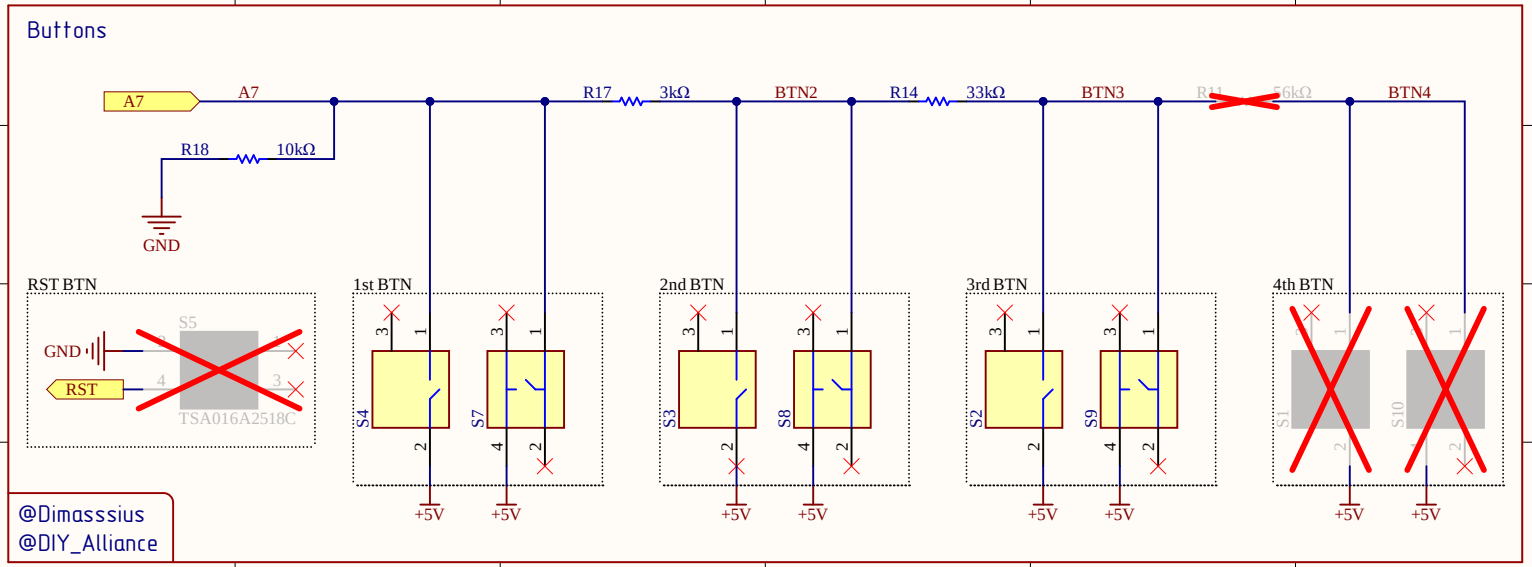


Anode Dots Resistor

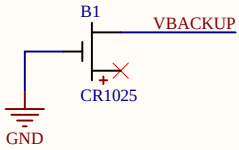


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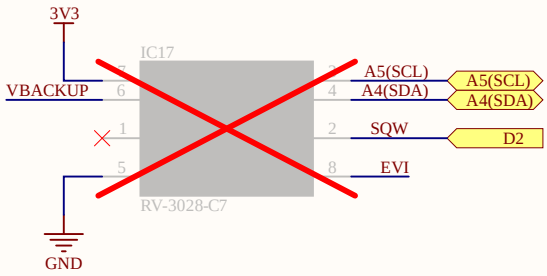

IO Port Expansion
IC9
74HC595D
TUBE_2
TUBE_3
TUBE_4
TUBE_5
TUBE_6
DPR
DPL
1 Q1
2 Q2
3 Q3
4 Q4
5 Q5
6 Q6
7 Q7
8 GND
16 VCC
15 Q0
14 DS
13 OE
12 STCP
11 SHCP
10 MR
9 Q7S
TUBE_1
D11(MOSI)
D0(RX)
D13(SCK)
+5V
+5V
GND
GND
@Dimasssius
@DIY_Alliance


Buttons
A7
A7
R18
10kΩ
GND
R17
3kΩ
BTN2
R14
33kΩ
BTN3
R11
56kΩ
BTN4
RST BTN
GND
RST
S5
TSA016A2518C
1st BTN
S4
S7
2nd BTN
S3
S8
3rd BTN
S2
S9
4th BTN
S1
S10
+5V
@Dimasssius
@DIY_Alliance

Battery Holder

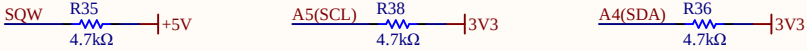


RV-3028-C7 - Variant #1

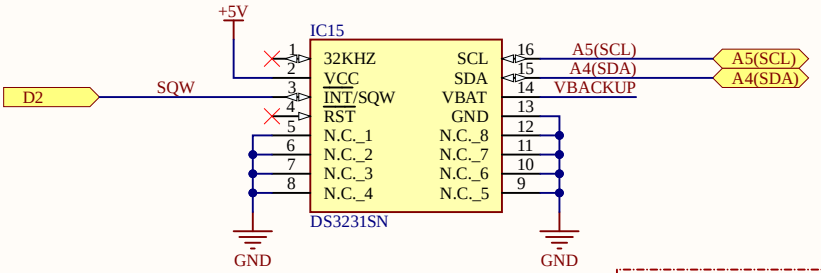




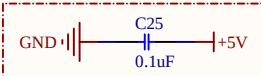
Pull-Ups

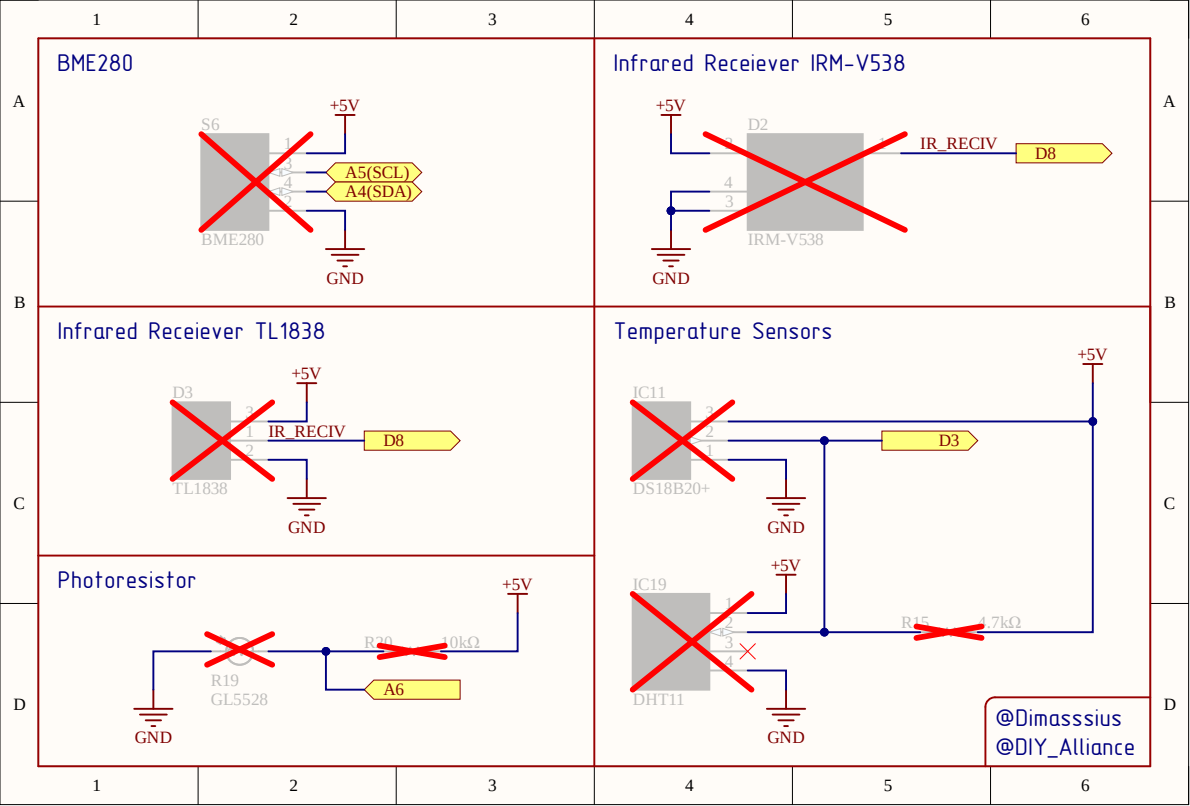


DS3231 - Variant #2



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@DIY_Alliance

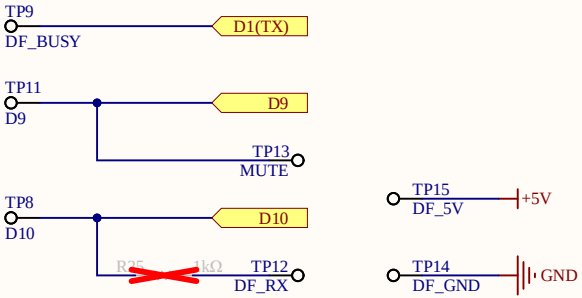



1
2
3
4
5
6
A
B
C
D
BME280
S6
+5V
A5(SCL)
A4(SDA)
BME280
GND
Infrared Receiever IRM-V538
+5V
D2
IR_RECIV
D8
IRM-V538
GND
Infrared Receiever TL1838
D3
+5V
IR_RECIV
D8
TL1838
GND
Temperature Sensors
+5V
IC11
D3
DS18B20+
GND
+5V
IC19
R15
4.7kΩ
DHT11
GND
Photoresistor
+5V
R20
10kΩ
R19
GL5528
A6
GND
@Dimasssius
@DIY_Alliance

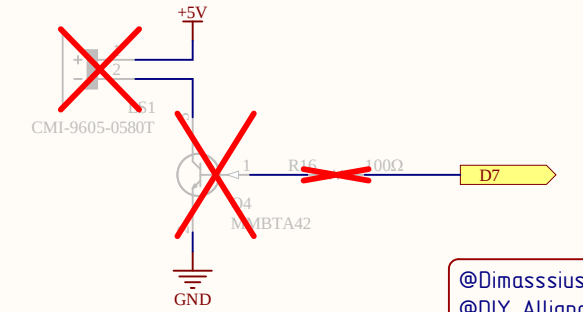
RDA5807M - Radio Tuner Connection Points



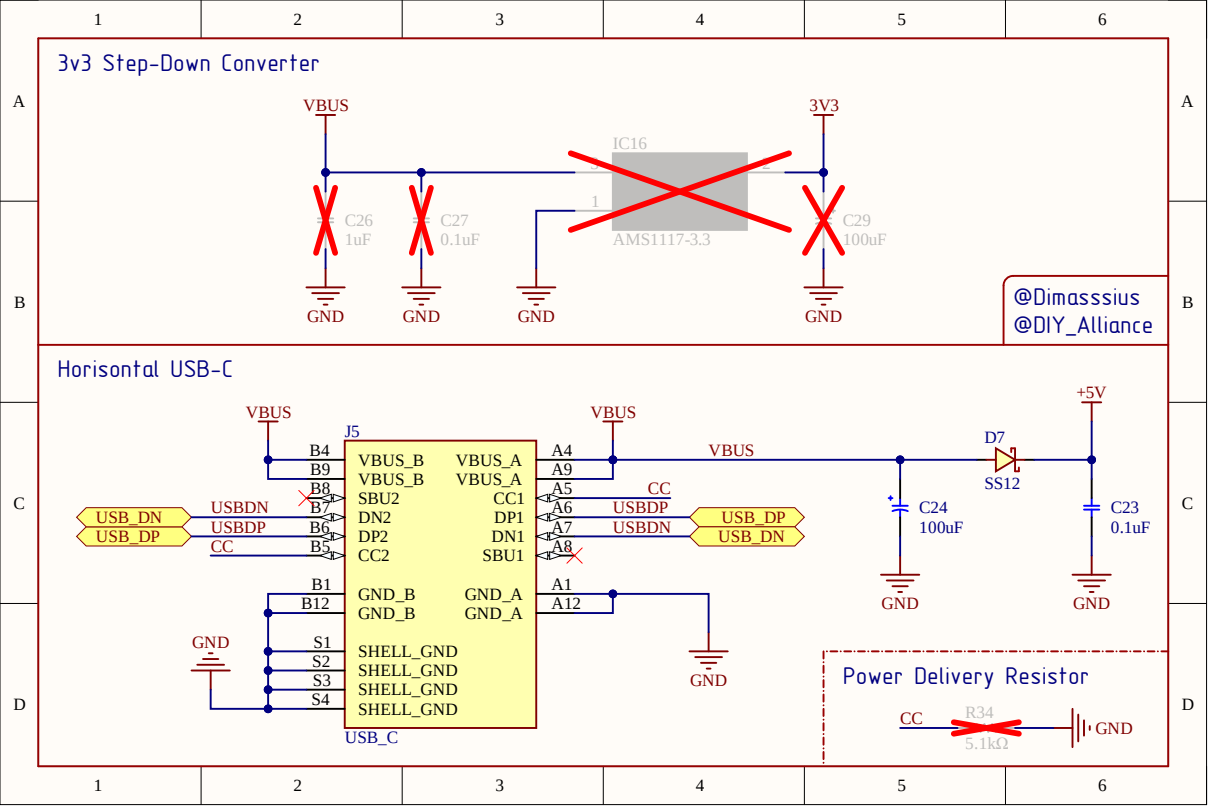
DFPlayer Connection Points

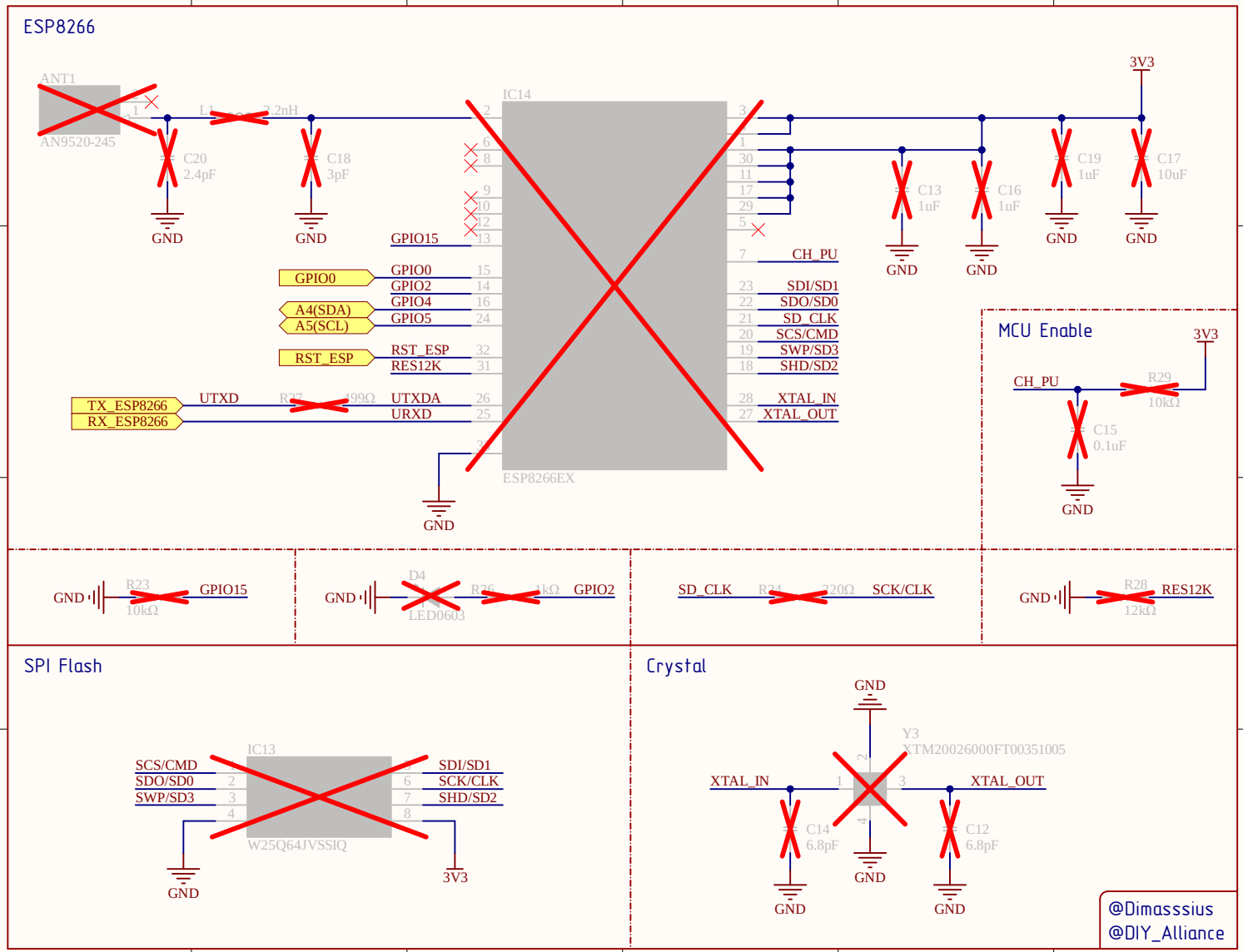


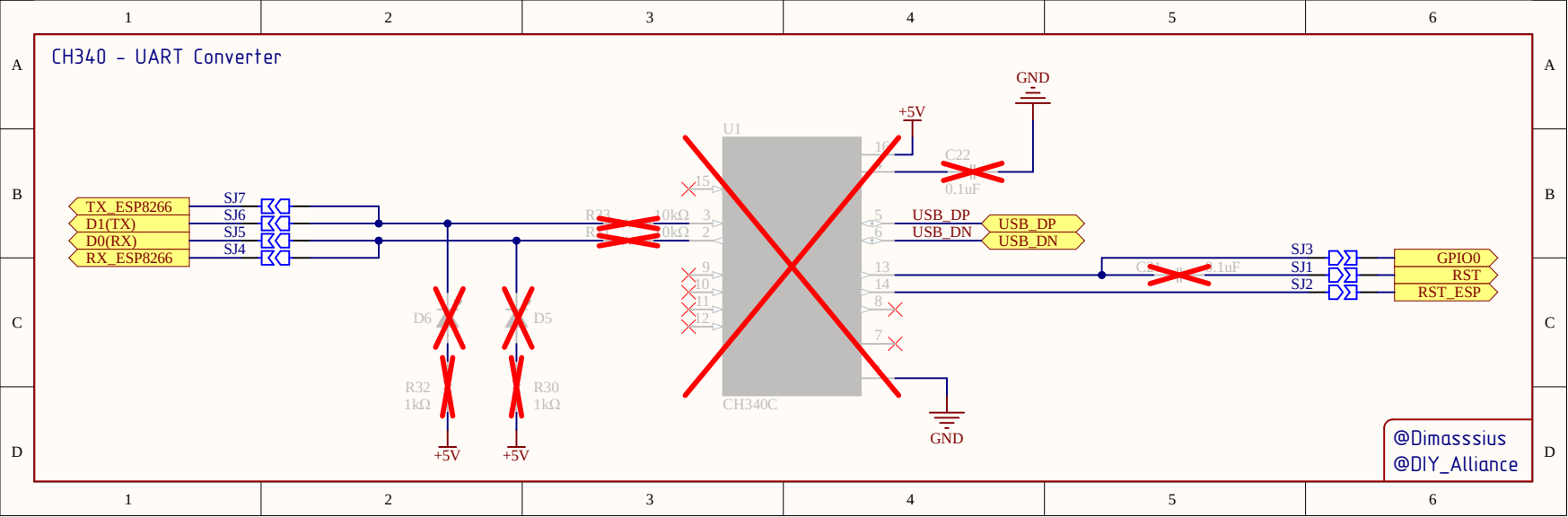
Buzzer



@Dimasssius
@DIY_Alliance


3v3 Step-Down Converter
VBUS
3V3
IC16
C26
1uF
C27
0.1uF
AMS1117-3.3
C29
100uF
GND
@Dimasssius
@DIY_Alliance
Horisontal USB-C
J5
USB_C
VBUS_B
VBUS_A
SBU2
CC1
DN2
DP1
DP2
DN1
CC2
SBU1
GND_B
GND_A
SHELL_GND
USB_DN
USB_DP
USBDN
USBDP
CC
+5V
D7
SS12
C24
100uF
C23
0.1uF
Power Delivery Resistor
R34
5.1kΩ


ESP8266
ANT1
AN9520-245
C20
2.4pF
C18
3pF
IC14
ESP8266EX
GPIO15
GPIO0
GPIO2
GPIO4
GPIO5
RST_ESP
RES12K
UTXDA
URXD
UTXD
TX_ESP8266
RX_ESP8266
CH_PU
SDI/SD1
SDO/SD0
SD_CLK
SCS/CMD
SWP/SD3
SHD/SD2
XTAL_IN
XTAL_OUT
3V3
C13
1uF
C16
1uF
C19
1uF
C17
10uF
GND
MCU Enable
R29
10kΩ
C15
0.1uF
R23
10kΩ
GPIO15
D4
LED0603
R26
1kΩ
GPIO2
SD_CLK
R24
220Ω
SCK/CLK
R28
12kΩ
RES12K
SPI Flash
IC13
W25Q64JVSSIQ
SCS/CMD
SDO/SD0
SWP/SD3
SDI/SD1
SCK/CLK
SHD/SD2
Crystal
Y3
XTM20026000FT00351005
C14
6.8pF
C12
6.8pF
@Dimasssius
@DIY_Alliance


CH340 - UART Converter
TX_ESP8266
D1(TX)
D0(RX)
RX_ESP8266
SJ7
SJ6
SJ5
SJ4
R33
10kΩ
R31
10kΩ
D6
D5
R32
1kΩ
R30
1kΩ
+5V
+5V
U1
CH340C
GND
+5V
C22
0.1uF
USB_DP
USB_DN
USB_DP
USB_DN
SJ3
SJ1
SJ2
GPIO0
RST
RST_ESP
GND
@Dimasssius
@DIY_Alliance